

ECO-SYN DEFORESTATION & ILLEGAL LOGGING DETECTOR



CMR COLLEGE OF ENGINEERING & TECHNOLOGY
(Autonomous)
Kandlakoya, Medchal, HYDERABAD



CMR THINKFEST 3.0
01st August 2026

TEAM ID-TFT002

NAME OF THE SDG(15)

Organized By
CENTRE FOR ENGINEERING EDUCATION RESEARCH

- **Describe the problem that you are trying to solve?**

Eco-Sync stops silent forest destruction by turning remote canopies into 24/7 intelligent listening networks that detect chainsaw threats on the edge and alert rangers instantly—no internet required.

- **Describe in detail the solution you will provide to solve the problem. What is your idea (App, Product, or Service) and how does it overcome a problem for your user?**

Eco-Sync transforms silent, endangered forest canopies into an intelligent listening network that catches illegal deforestation in real time. By running local edge AI directly on tree-mounted listening nodes, the system analyzes ambient forest acoustics to detect chainsaw and gunshot sounds within seconds without sending raw audio to the microcontroller. The moment a threat is verified, it transmits a tiny location-tagged alert over long-range LoRa radio to a central gateway, which immediately routes an SMS with exact GPS coordinates straight to a ranger's phone—stopping illegal logging before the trees ever hit the ground.

1. Customer Segments

- **For whom are we creating value?**

We are giving underfunded forest rangers, conservation organizations, and frontline indigenous communities a real-time superpower over silent environmental destruction. By replacing blind, exhausted foot patrols with precise acoustic intelligence, Eco-Sync transforms frontline guardians from helpless witnesses into instant responders—shielding Earth's most critical biodiversity hotspots before the damage is done.

- **Who are our most important customers?**

Our most critical customers are government forestry departments and national park authorities tasked with protecting vast, understaffed reserves. Alongside them, global conservation NGOs and indigenous forest guardians are vital adopters who urgently need low-cost, off-grid intelligence to stop illegal logging and safeguard biodiversity.

2. Value Proposition

- **What value do we deliver to our customers?**

We give forest guardians the ultimate tactical advantage: eyes and ears where internet cannot reach. By turning raw canopy sounds into instant, GPS-precise threat alerts, Eco-Sync transforms delayed disaster management into proactive, zero-blindspot interception—saving thousands of acres on a fraction of a traditional budget.

- **Which customer problems are we helping to solve?**

We shatter the zero-coverage blind spots, delayed threat alerts, and astronomical patrol costs that leave forest guardians helpless against silent, rapid logging

- **What bundles of products and services are we offering to each customer segment?**

We offer ready-to-install listening hardware, long-range signal receivers, and automated alert software, backed by training and technical support for every deployment team.

- **Which customer needs are we satisfying?**

We satisfy the critical need for instant, off-grid threat visibility and precise GPS alert routing so underfunded guardians can stop illegal logging without cellular coverage or costly manual patrols

3. Key Activities

- **Which key activities do our value propositions require?**

Manufacturing weather-proof canopy nodes, continuously training edge AI acoustic models, and ensuring ultra-low power hardware reliability.

- **Which key activities do our distribution channels require?**

Partnering with regional conservation networks, facilitating on-site pilot deployments, and managing international hardware logistics.

- **Which key activities do our customer relationships require?**

Providing hands-on ranger onboarding, offering 24/7 technical support, and maintaining active feedback loops with frontline teams.

- **Which key activities do our revenue streams require?**

Structuring hardware-plus-subscription contracts, managing grant funding pipelines, and processing software licensing renewals.

4. Customer Relationships

- **Which type of relationship does each customer segment expect us to establish and maintain?**

High-touch, collaborative co-creation with dedicated technical support and hands-on ranger training tailored to remote operational realities.

- **Which relationships have we already established?**

Close pilot partnerships with local forestry authorities, regional conservation NGOs, and select indigenous land management groups.

- **How are they integrated with the rest of our business model?**

Feedback from frontline rangers feeds directly into R&D for software updates, while successful conservation outcomes generate field data to secure grant funding and new sales.

- **How costly are they?**

Moderately high upfront during pilot setup and field onboarding, but shifting to low-cost, automated remote maintenance and self-service platform management over time.

5. Revenue Streams

- **For what value are our customers really willing to pay?**

Early-warning threat detection and actionable GPS intelligence that actually stops illegal logging and proves quantifiable acreage protection to stakeholders.

- **For what do they currently pay?**

High-friction manual foot patrols, satellite imagery with multi-day reporting delays, and expensive heavy surveillance equipment that fails off-grid.

- **How are they currently paying?**

Upfront capital expenditures funded through annual government budgets, international conservation grants, and multi-year donor allocations.

- **How much would they prefer to pay?**

Low-risk, predictable annual subscriptions or operational grants that bundle hardware, maintenance, and software updates into one transparent fee.

- **How much does each revenue stream contribute to the overall revenue?**

Hardware node sales and deployment kits contribute roughly 60% of overall revenue, while annual software licensing and maintenance support contribute the remaining 40%.

6. Key Resources

- **What key resources do our value propositions require?**

Edge AI acoustic algorithms, ruggedised canopy hardware, and specialized embedded-systems engineers.

- **What key resources do our distribution channels require?**

Regional conservation partnerships, international logistics networks, and field demonstration kits for pilot testing.

- **What key resources do our customer relationships require?**

Dedicated multilingual field-support staff, comprehensive training manuals, and an accessible remote customer portal.

- **What key resources do our revenue streams require?**

Automated billing systems, grant writing specialists, and secure contract management platforms.

7. Key Activities

- **Which key activities do our value propositions require?**

Manufacturing weather-proof canopy nodes, continuously training the edge AI acoustic models, and ensuring ultra-low power hardware reliability.

- **Which key activities do our distribution channels require?**

Partnering with regional conservation networks, facilitating on-site pilot deployments, and managing international hardware logistics.

- **Which key activities do our customer relationships require?**

Providing hands-on ranger onboarding, offering 24/7 technical support, and maintaining active feedback loops with frontline teams.

- **Which key activities do our revenue streams require?**

Structuring hardware-plus-subscription contracts, managing grant funding pipelines, and processing software licensing renewals.

8. Key Partners

- **Who are our key partners?**

Government forestry agencies, international conservation NGOs (such as WWF and Rainforest Alliance), and indigenous community forest associations.

- **Who are our key suppliers?**

Specialized acoustic sensor manufacturers, industrial-grade microchip vendors, and solar panel/battery component suppliers.

- **Which key resources are we acquiring from our partners?**

Specialized acoustic sensor manufacturers, industrial-grade microchip vendors, and solar panel/battery component suppliers.

- **Which key activities do our partners perform?**

Assisting with physical device deployment, conducting frontline ranger patrols based on system alerts, and securing institutional funding.

9. Cost Structure

- **What are the most important costs inherent in our business model?**

Hardware manufacturing and assembly, R&D for acoustic AI algorithms, and international field deployment and ranger onboarding logistics.

- **Which key resources are the most expensive?**

Specialized embedded-systems and AI engineering talent, alongside custom weatherproof electronic hardware and microchips.

- **Which key activities are the most expensive?**

On-site pilot deployments and field testing, continuous training and fine-tuning of machine learning acoustic models, and global logistics for delivering hardware to remote regions.

*Thank
you!*